13. The action by mycopesticide depends on the
    a. Ingestion of fungi by insects
    b. Penetration of cuticle by fungi
    c. Ingestion of leaves infected with fungi
    d. Ingestion of spores by fungi.
14. Crown gall is caused by *Agrobacterium* on
    a. Monocots
    b. Dicots only
    c. Monocots and dicots
    d. Ferns
15. Smut is a disease caused by
    a. *Puccinia sp*
    b. *Ustilago sp*
    c. *Verticillium*
    d. None of the above

## Answer the following

1. What is the role of purple and green bacteria in sulphur cycle?
2. What is nitrogenase? Give its function.
3. Define biofertilizers.
4. What is VAM?
5. Define biopesticides.
6. What is a smut?
7. What is NPV?
8. List out Bacterial diseases of plants.
9. What is the end result of decomposition of organic matter in carbon cycle? Give the role of microorganisms with examples.
10. What is the function of leghemo-globin?
11. Give the method of application of *Azolla* in paddy fields.
12. Give the salient features of *Rhizobium*.
13. Explain the mass production of phosphate solubilisers.
14. What is *Azolla-Anabaena* symbiosis?
15. What is crystal toxin? Write short notes on BT cotton.
16. Explain the process of root nodule formation with appropriate diagram.
17. Give in detail the reactions involved in phosphorus/carbon/sulphur/nitrogen cycle.
18. Explain the mass production of VAM.
19. Give detailed account on *Bacillus thuringiensis*.

### Student Activity

- Sow two groundnut seeds in a plastic cup/earthern pot. After one month, pull out the plant and observe the root system for nodules.
- Collect pictures of all organisms involved in sulphur cycle and prepare a collage showing its role.
- Prepare a chart work showing the biogeochemical cycles of carbon, nitrogen and phosphorus
- Collect pictures of various diseases of plants and prepare a chart.
- Prepare a model on the mode of action of BT.
- Collect diseased parts of plants and identify the symptoms of the disease.



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